T24BCD059

Department of Computer Engineering
Faculty of Engineering and Technology
Jamia Millia Islamia, New Delhi

Academic Session 2025-2026
1st Mid Semester Examination B.Tech. (Computer Science & Engineering) Data Science 4th Sem

SUBJECT: DSC-401 **Data Mining** **Credits =3**

TIME: 01Hours **Max. Marks: 12**

Note:

1. *M-Marks, L-BloomsLevel, C–CourseOutcome*
2. *Electronic gadgets are not allowed in the examination hall like mobile phone, smart watch etc.*
3. *Students will only be allowed to use non-programmable scientific calculator.*
4. **All questions are compulsory**

S.No.	Questions	M	L	C
1.	What are the necessary conditions for a problem to be solved using classification techniques.	2	2	1
2.	Which central tendency measure is well suited for "assignment submission delay (in days)". Justify and calculate.	2	3	1
3.	A hospital is developing a learning system to identify patients with similar medical profiles in order to recommend appropriate treatments. The system represents each patient using a feature vector of laboratory test values. The following three laboratory measurements were conducted on two patients:	4		1

Test	Patient A	Patient B	Patient C
White Blood Cell Count (WBC)	10	15	100
C-Reactive Protein Level (CRP)	20	30	200
Inflammation Marker Score (IMS)	30	45	300

A data scientist suggests using *cosine similarity* to determine whether the two patients have similar medical profiles.

(A) Represent the medical profiles of the patients as vectors and compute the cosine similarity between Patients A and B and between Patients A and C. — L: 4

(B) Clinically interpret the similarity values obtained for the medical profiles of the patients. Does cosine similarity yield useful results in this case? If not, suggest a better method for identifying similar medical profiles. — L: 5

4.	For the following dataset, with minimum support = 50%.	4	3	2

TID	Items
T1	A, B, C
T2	A, C, D
T3	B, D
T4	A, B, C
T5	B, C, E

Use FP-Growth algorithm to construct FP-tree. Show each step clearly.

Bloom's Taxonomy Levels (1-Remembering, 2-Understanding, 3- Applying, 4- Analyzing, 5- Evaluating, 6- Creating)